Product Requirement Document

Team Name
git force push

Domain 2
AI for Economic Empowerment & Decision Intelligence

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1. Project Overview

- **Problem Statement:** SME merchants make high-stakes daily decisions, what to restock, what to charge, when to hire, entirely on intuition. They have no access to the kind of market intelligence that large chains take for granted. A Starbucks regional manager knows a drought in Colombia will raise coffee prices next month. The owner of a kedai kopi in Subang does not. Meanwhile, supply chain disruptions are increasing, extreme weather events are affecting logistics, and rising costs are squeezing already thin margins. Without real-time intelligence, most SME owners only recognise risks and opportunities when it is already too late.

- **Target Domain:** AI-powered decision intelligence for small and medium enterprises in Southeast Asia — covering inventory management, market trend detection, location intelligence, and operational planning.

- **Proposed Solution Summary:** BijakSense is an AI-powered decision co-pilot that continuously monitors global and local signals — commodity prices, weather events, supply chain disruptions, regional consumer trends — and translates them into plain-language business decisions. Not raw data, not charts. Just: what's happening, what it means for your business, and what to do about it. The merchant shifts from being overwhelmed by the world to being one step ahead of it.

2. Background & Business Objective

- **Background of the Problem:** For years, business intelligence has been the exclusive domain of large enterprises with dedicated analysts, procurement teams, and expensive software. An SME merchant running a bubble tea stall in Petaling Jaya makes the same categories of decisions as a regional chain manager, restock timing, pricing strategy, location expansion, staffing levels, but with none of the same infrastructure. They rely on gut feel, supplier gossip, and lagging indicators like empty shelves or declining sales. By the time they notice a problem, the opportunity to act has already passed.

- **Importance of Solving This Issue:** SMEs make up over 99% of all businesses in Southeast Asia according to the ASEAN SME Policy Index. Improving the decision-making quality of even a fraction of these businesses has outsized economic impact. Reducing a single bad restocking decision per month, or helping a merchant spot one trending product category early, can meaningfully improve profitability on margins that are already thin. At scale, BijakSense democratises access to the kind of intelligence that has historically only been available to the top 1% of businesses by size.

- **Strategic Fit / Impact:** BijakSense directly addresses the UMHackathon Domain 2 mandate: using Z.AI's GLM to analyse information, generate insights, and support decision-making that improves real-world outcomes. The multi-agent architecture powered by GLM is not an integration layer — it is the core reasoning engine that makes the world-signal-to-decision translation possible.

3. Product Purpose

3.1 Main Goal of the System: To give every SME merchant in Southeast Asia access to the same quality of business intelligence that large enterprises take for granted — delivered as plain-language decisions, in real time, personalised to their specific business.

3.2 Intended Users (Target Audience):

- F&B stall and restaurant owners
- Retail shop operators
- Service business owners
- First-time business founders in the pre-launch phase
- SME support organisations and incubators serving large merchant cohorts

4. System Functionalities

4.1 Description: BijakSense operates as a multi-agent intelligence system. Four specialist AI agents, each powered by Z.AI's GLM continuously monitor different streams of world data. A central Orchestrator agent routes incoming signals to the appropriate specialist, collects structured JSON outputs, and synthesises them into a unified merchant-facing recommendation. The merchant interacts with the system through a live intelligence map, a prioritised dashboard, and a conversational chat interface.

4.2 Key Functionalities:

- **Intelligence Map:** A live Southeast Asia map where world signals appear as pulsing dots by urgency colour. When a signal is relevant to the merchant, an animated arc draws from the signal origin to the merchant's location pin. Merchants can toggle between regional and local 5km views. Clicking any signal dot opens a detail panel showing the signal summary, business impact category, and source.
- **Command Center (Dashboard):** The merchant's morning briefing. Three summary stat cards (needs attention, supply risk items, resolved today). Below them, a prioritised list of

AI insight cards — each showing the responsible agent, signal source, and a one-line plain-English action. No charts, no numbers to decode. Just ranked decisions.

- **AI Chat (Co-Pilot):** Conversational interface powered by GLM. The merchant asks anything in natural language. The Orchestrator routes the query to the relevant specialist agents and returns a specific, reasoned answer with attribution to the agent that produced it.
- **Inventory Planner:** Deep-dive view showing current stock levels per ingredient, restock recommendations with quantity and cost estimates, and price trend forecasts. Triggered by commodity and weather signals.
- **Market Analyst:** Trend radar showing emerging vs. fading product categories in the merchant's area, competitor pricing benchmarks, and specific product and pricing recommendations. Triggered by consumer trend and competitor signals.

4.3 AI Model & Prompt Design

4.3.1 Model Selection: Z.AI's GLM is used as the primary reasoning engine for all four specialist agents and the Orchestrator. GLM was selected for three reasons: it is the required model specified by the domain problem statement; it supports structured JSON output which is essential for agent-to-agent communication via the A2A protocol; and it provides strong multilingual reasoning capability, which is important for a Southeast Asian merchant context where signals and queries may mix Bahasa Malaysia and English.

4.3.2 Prompting Strategy: BijakSense uses a multi-step agentic prompting strategy. Each GLM invocation is composed of four structured blocks: an agent identity block defining the agent's domain and output schema, a merchant context block containing the merchant's profile and ingredient dependencies, a signal block containing the world event being processed, and an instruction block specifying the exact task. This approach was chosen over zero-shot prompting because SME decision recommendations require grounding in specific merchant context — a generic prompt about a typhoon produces a generic answer, while a context-rich prompt produces "restock 60kg of jasmine rice within 5 days at RM 156." Few-shot examples are embedded in the agent identity blocks to reinforce the output format.

4.3.3 Context & Input Handling: Each agent receives a structured context payload assembled by the prompt builder module. The merchant profile block is capped at approximately 800 tokens. The signal block is capped at 300 tokens. The conversation history passed to the chat Orchestrator is limited to the last 10 exchanges, with older messages summarised into a single context paragraph. Inputs that exceed these limits are chunked, with the most recent and most relevant content prioritised. For chat queries exceeding 500 tokens, the system extracts the core question and passes only that to GLM, logging the full input separately.

4.3.4 Fallback & Failure Behaviour: If GLM returns a response that fails JSON schema validation, the agent retries once with a stricter prompt appended: "You must respond with valid JSON only matching the exact schema provided. No preamble, no explanation." If the retry also fails validation, the system logs the failure, skips saving the recommendation, and surfaces a graceful error state on the dashboard: "This insight is temporarily unavailable — we are retrying." For the chat interface, if GLM returns an off-topic or unusable response, the Orchestrator detects low relevance via keyword matching against the merchant's profile and responds with: "I couldn't find a confident answer for that. Try rephrasing or ask me about your inventory, pricing, or local signals."

5. User Stories & Use Cases

User Stories:

- As an SME merchant, I want to open the app each morning and immediately see what decisions need my attention today, so I do not need to read the news or analyse data myself.
- As a merchant, I want to be alerted when a world event will affect my ingredient costs before the price change hits my supplier invoice, so I can restock in advance.
- As a merchant, I want to ask the system a question in plain language and get a specific answer, so I do not need to interpret raw data or charts.
- As a first-time founder, I want to see which locations near me have high foot traffic and low competitor density, so I can make an informed decision about where to open.
- As a merchant, I want to know which products are trending in my area before my competitors act on them, so I can adjust my menu first.

Use Cases (Main Interactions):

- **Signal-to-recommendation flow:** A commodity price API detects palm oil futures up 11%. The relevance filter scores this signal against the merchant's ingredient list — high match. The Orchestrator routes it to the Inventory Planner agent. GLM reasons over the signal and merchant context, produces a structured JSON recommendation. The recommendation surfaces on the Command Center as an amber insight card: "Check creamer and syrup margin impact and adjust pricing before the next reorder cycle."
- **Chat query flow:** The merchant types "Should I raise my matcha price?" The Orchestrator passes the query, merchant profile, and the last 5 active recommendations to GLM. GLM reasons over competitor pricing data and the current matcha trend signal and returns: "Your matcha milk tea at RM 10.00 is 15% below the area average of RM 11.80. With matcha trending +38% regionally, demand is strong enough to support a RM 0.50–1.00 increase without losing customers."

- **Location scout flow:** The merchant switches the Intelligence Map to Local 5KM view. The AI Expansion Scout panel appears. They set business type, expansion goal and max rent. The Location Scout agent scores candidate sites within the radius and pins gold markers on the map. Clicking a marker opens a recommendation panel with location score, foot traffic estimate, rent estimate, and key advantages.

6. Features Included (Scope Definition)

- Live Intelligence Map with Southeast Asia regional and local 5KM views, signal dots, animated arcs, and signal detail panels
- Command Center dashboard with priority-ranked AI insight cards and summary stat cards
- AI Chat interface with real GLM API calls and Orchestrator routing
- Inventory Planner deep-dive with stock level indicators and restock recommendations
- Market Analyst deep-dive with trend radar and competitor pricing benchmarks
- AI Expansion Scout on the Intelligence Map for location scouting within a configurable radius
- Multi-agent A2A architecture with four specialist agents and one Orchestrator
- MCP-compatible tool interface placeholders for all four agents
- Demo data seeded for a bubble tea merchant in Petaling Jaya covering all five screens

7. Features Not Included (Scope Control)

- Live production data feeds (signals are simulated for demo purposes)
- POS system integration or real sales data ingestion
- Payment systems or financial forecasting tools
- Real open banking integration
- Mobile app (web only for this submission)
- Multi-merchant accounts or team collaboration features
- Automated execution (e.g. auto-placing supplier orders from a recommendation)

8. Assumptions & Constraints

LLM Cost Constraint: Estimated token cost per average user session is approximately 8,000–12,000 tokens, covering onboarding context, 3–5 agent signal processings, and 2–3 chat exchanges. To keep inference costs manageable, agent outputs are cached in Redis with a 2-hour TTL — if the same signal is processed for a merchant within the cache window, the cached recommendation is returned without a new GLM call. The merchant context block is also pre-built and cached per merchant so it is not recomputed on every agent invocation.

Technical Constraints: Signal data is simulated for the demo. Real integration with commodity APIs, weather APIs, and news feeds is partially implemented but not fully connected in the submitted version. The MCP tool interfaces for all four agents are defined as placeholder schemas — full MCP handler implementation is out of scope for the hackathon timeline.

Performance Constraints: GLM API response time varies between 2–5 seconds per call. For the chat interface, if a response exceeds 5 seconds, the system streams the response token by token using FastAPI's StreamingResponse to avoid a blank wait state. Dashboard load is designed to complete under 200ms by reading only from PostgreSQL and Redis — no GLM calls on dashboard load.

User Input: The system requires an initial merchant profile (sector, location, product list) to produce personalised recommendations. Without this profile, all recommendations default to generic category-level signals. The system is not designed for production deployment without human review of recommendations — it is a decision support tool, not an autonomous decision-maker.

9. Risks & Questions Throughout Development

Signal Relevance Accuracy: How do we ensure that the relevance filter correctly matches world signals to the right merchants without producing irrelevant alerts? Current mitigation: scoring logic uses ingredient keyword matching, geographic proximity via haversine distance, and category adjacency rules — but edge cases remain.

GLM Output Consistency: How do we ensure GLM produces consistent structured JSON across different signal types and merchant profiles? Current mitigation: JSON schema validation with one automatic retry, but hallucinated field values (e.g. fabricated price figures) cannot always be detected programmatically.

Demo Stability: How do we ensure the live demo runs without a broken state in front of judges? Current mitigation: all five demo screens are seeded with pre-generated data as fallback. If a live GLM call fails during the demo, the fallback data renders seamlessly.

Trend Data Freshness: What happens when a trend signal is no longer accurate by the time the merchant acts on it? Current mitigation: all recommendations carry a generated timestamp and a 48-hour expiry. Expired recommendations are visually marked as stale on the dashboard.